

17a) The dislocations will repel each other.

b) The dislocations will attract each other and annihilate.

$$18) \tau_R = \sigma \cos \lambda \cos \phi$$

$$\phi = 28.1^\circ$$

To get the most favoured slip direction, need to maximise τ_R , hence maximising $\cos \lambda$.

$$\cos 62.4^\circ = 0.4632960351$$

$$\cos 72.0^\circ = 0.3090169944$$

$$\cos 81.1^\circ = 0.1547103863$$

$\therefore \lambda = 62.4^\circ$ is the most favoured

$$\begin{aligned} b) \tau_R &= 1.8 \cos 28.1^\circ \cos 62.4^\circ \\ &= 0.7356345831 \\ &\approx 0.74 \text{ MPa} \end{aligned}$$

$$1a) \sigma_y = \sigma_0 + kD^{-0.5}$$

Original yield strength:

$$\begin{aligned}\sigma_{yi} &= 185 + 40(0.5)^{-0.5} \\ &= 241.5685425 \\ &\approx 241.6 \text{ MPa}\end{aligned}$$

Yield strength after annealing:

$$d^2 - d_0^2 = kt$$

$$d^2 = 2 - 2(60 \times 60) + 0.5^2$$

$$d = 88.99578642$$

$$\approx 89 \mu\text{m}$$

$$\sigma_{yf} = 185 + 40(89)^{-0.5}$$

$$= 189.2400919$$

$$\approx 189.2 \text{ MPa}$$

$$\begin{aligned}
 20) \% CW &= \frac{A_o - A_d}{A_o} \times 100 \\
 &= \frac{\frac{\pi d_i^2}{4} - \frac{\pi d_f^2}{4}}{\frac{\pi d_i^2}{4}} \times 100 \\
 &= \frac{d_i^2 - d_f^2}{d_i^2} \times 100 \\
 &= \frac{10^2 - 7.5^2}{10^2} \times 100 \\
 &= 43.75\%
 \end{aligned}$$

For $\% CW = 43.75\%$,

$$\sigma_y = 420 \text{ MPa}$$

$$TS = 540 \text{ MPa} > 380 \text{ MPa}$$

$$\% EL = 6 < 15$$

$\therefore$ The required properties cannot be achieved with a single step.

20) For $TS > 380 \text{ MPa}$,
 $\% CW > 12\%$

For $\% EL > 15$,
 $\% CW < 27\%$

$$\therefore 12 < \% CW < 27$$

Picking the $\% CW$ to be 20% for the second cold work step,

$$\% CW = \frac{d_i^2 - d_f^2}{d_i^2} \times 100$$

$$20 = \left(1 - \left(\frac{d_f}{d_i}\right)^2\right) \times 100$$

$$0.2 = 1 - \left(\frac{d_f}{d_i}\right)^2$$

$$\left(\frac{d_f}{d_i}\right)^2 = 0.8$$

$$7.5^2 = 0.8 d_i^2$$

$$d_i = \frac{15\sqrt{5}}{4}$$

$$\approx 8.39 \text{ mm}$$

20) Step 1: Reduce diameter to 8.39mm

$$\begin{aligned}\%CW &= 1 - \left(\frac{8.39}{10}\right)^2 \\ &= \frac{19}{64} \\ &\approx 29.7\%\end{aligned}$$

Step 2: Anneal so that metal fully recrystallises.

Step 3: Reduce diameter to 7.5mm

$$\begin{aligned}\%CW &= 1 - \left(\frac{7.5}{8.39}\right)^2 \\ &= 20\%\end{aligned}$$

From the graphs:

$$\sigma_y = 340 \text{ MPa}$$

$$TS = 400 \text{ MPa}$$

$$\%EL = 24$$

$\therefore$ All criteria is satisfied.